

Few-Shot Classification with Multisemantic Information Fusion Network

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ABSTRACT

Metric-based methods aim to predict class labels by computing the similarity between samples using distance functions, which is the mainstream approach to few-shot learning. However, the limited representational space of feature vectors and appearance variations among congenetic samples still present challenges. We propose a Multisemantic Information Fusion Network (MIFN) to address these problems. A Lower-level Feature generator (LF-generator), which is an unsupervised module, adaptively activates high-response regions of objects to introduce discriminative semantic details. Meanwhile, a Higher-level Feature extractor (HF-extractor) learns global semantic information with human cognition to minimise the impact of appearance variations. We integrate the coarse outputs of these two modules, which complement each other to jointly promote more precise predictions. Furthermore, considering the importance of prototypes, we redefine the sampling strategy of the triplet loss and utilise it as an auxiliary loss to sharpen the decision boundary at the prototype level, facilitating subsequent classification. Our experimental results demonstrate the competitiveness of our approach in both general few-shot classification (*mini*-ImageNet and *tiered*-ImageNet) and cross-domain problems (CUB, Caltech-101, Stanford-dogs, and Stanford-cars) with minimal bells and whistles.

1. Introduction

The traditional convolutional neural network achieves excellent performance in specific tasks with abundant labelled samples[6, 22]. However, the reliance on large annotated datasets and the limited generalisation ability of these models are obstacles. Motivated by human learning activities, few-shot learning, which adapts quickly to novel tasks with only a small number of labelled samples, has emerged.

Metric-based method is a prevalent branch of few-shot learning, in which the similarity between feature vectors in the learned embedding space is used to predict class labels. Hence, many methods in this field focus on either choice of measurements or learning more ideal embedding space. In terms of distance function, Matching network[40] and Prototypical network[34] (ProtoNet) utilised cosine similarity and Euclidean distance, respectively, as simple measurements. Relation network[36] surpassed traditional distance functions by leveraging convolutional neural networks. Zhang et al.[48] introduced the Earth-Mover distance to treat few-shot classification as a linear programming problem, using the optimal matching cost to represent similarity. However, vector representations omit semantic information that is crucial for the distance function. Therefore, researchers must consider how to enhance vectorial representations to carry richer information, i.e., learning both discriminative and holistic feature vectors. Gidaris and Komodakis[12] utilised past knowledge learned from base classes with an attention mechanism to generate novel vector representations. Hou et al.[15] highlighted object regions using cross-attention maps to make the features discriminative. Ouali et al.[27] leveraged contrastive learning to capture more transferable information. Wertheimer et al.[44] reconstructed feature maps to store more information while making the homology more similar. The above methods primarily focus on seeking information within the intra-class inter-set to extract features that represent the class, but this introduces the noise between sets. In contrast, we strive for discriminative semantic information in the inter-class intra-set, emphasising the intrinsic differentiation of each class. Further, we incorporate higher-level global representations with human recognition into few-shot classification, an aspect that has not been explored in previous

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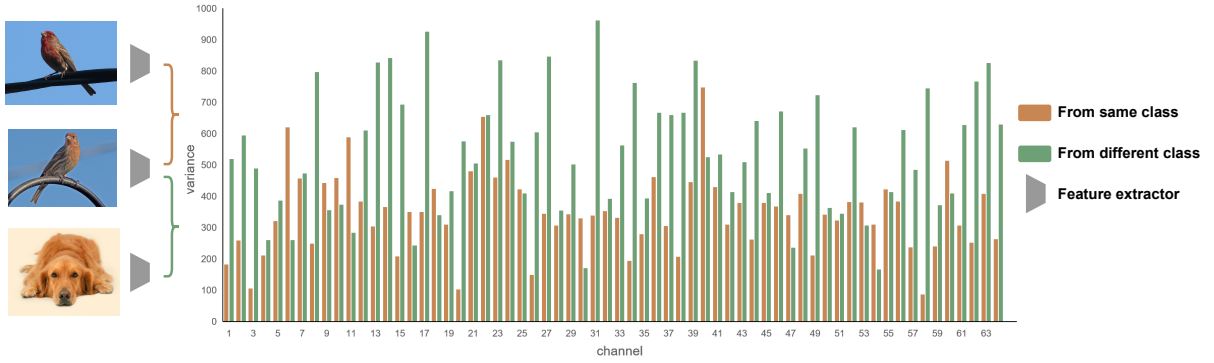


Figure 1: The variances of channels. We calculate the variances of the feature channels from the same classes and different classes and illustrate the sum of the variances per 10 channels. The majority of channel variances from heterogeneous samples (green) are significantly higher than those from congenetic samples (orange). This shows that the variances are highly responsive to the discriminative channels.

studies. This helps overcome appearance variations and indirectly ensures consistency among samples from the same class.

We propose a Multisemantic Information Fusion Network (MIFN) to generate feature vectors with more comprehensive semantic information for few-shot classification, comprising two streams: a Lower-level Feature generator (LF-generator) and a Higher-level Feature extractor (HF-extractor). Spp-Net[13] showed that the filters of deep convolutional layers have a high response to specific semantic content, and Xu et al.[45] treated the channels of feature maps as part detectors to fix specific information. Considering the above methods, we recognise the significance of the relationship between representations and semantic information. As shown in Fig.1, the summed variances per 10 channels from the two congeneric samples are generally lower than that of the inhomogeneity. This indicates that the responses of different channels to a certain category are disparate, and the channel variance implicitly reflects the semantic content of the corresponding class. Given the features extracted from the support set, we present that LF-generator utilises variances for selecting channels with more discriminative semantic information as activators for each feature. Without any label supervision, the activators inject high-response content into the corresponding feature vector. In addition to local discriminative features, few-shot classification values global semantic information. Since appearance variations can occur in homologous samples, relying solely on discriminative semantic information may lead to confusion. Therefore, we introduce an HF-extractor to learn higher-level global semantic information, which simulates human understanding and helps the model discern the overall semantic shape of the objects. Finally, the outputs of the two streams are coarse and complementary, each aiming to highlight discriminative and holistic semantic information, respectively. We fuse them together to mutually refine, producing a more precise result.

To obtain high-quality prototypes for simplifying subsequent induction inference, we introduced triplet loss at the prototype level as an auxiliary loss function. The classification criteria of ProtoNet relies on the mean of each class in the support set, to wit, prototypes. Rough prototypes lead to confusion in predictions. Therefore, obtaining more representative prototypes results in clear decision boundaries and, consequently, more accurate results. We aim to widen the gap between the otherguess prototypes by incorporating additional loss constraints, which can indirectly improve the separability of classes. Regarding the auxiliary loss, we developed an appropriate sampling strategy. The prototypes naturally constitute the positive pairs for each associated sample. We calculated the similarity between the prototypes of each class and selected the most similar prototypes from different classes to form the negative pair for the relevant class samples.

Our method can be seamlessly integrated into ProtoNet without the supervision of class information to improve performance. We demonstrate the efficacy of the proposed method in a single domain using *mini*-ImageNet[40] and *tiered*-ImageNet[29]. For cross-domain scenarios, we validate the generalisation ability for novel classes on the CUB[41], Caltech-101[9, 10], Stanford-dogs[18], and Stanford-cars[19] datasets. Note that we do not employ any additional techniques apart from pre-training in the training.

In summary, our contributions are as follows:

- 1) We propose a Multisemantic Information Fusion Network (MIFN) in which two types of semantic information complement and rectify each other. This network can be seamlessly embedded into ProtoNet to augment feature vectors.
- 2) We introduce two novel modules, namely the Lower-level Feature generator (LF-generator) and Higher-level Feature extractor (HF-extractor), for unsupervised generation of discriminative semantic details without any label supervision in the intra-set inter-class and learning higher-level semantic information mimicking human cognition of objects, respectively.
- 3) We reformulate the sampling strategy of the triplet loss as an auxiliary loss to separate prototypes for subsequent classification.
- 4) Our method achieves excellent performance in general few-shot classification and cross-domain problems.

The remainder of this paper is organised as follows. In Section 2, we review the work related to few-shot learning and develop a comprehensive understanding of the issues that need to be addressed. Section 3 describes in detail the implementation and fusion of our proposed methods, the LF-generator and HF-extractor, and defines the resampling strategy for Triplet loss. In Section 4, we report several comparative experiments that were conducted to verify the competitiveness of our method. Section 5 describes several ablation experiments conducted to analyse the effectiveness of the proposed methods. In Section 6, we conclude our work.

2. Related work

Few-shot learning trains many tasks, each consisting of a finite number of samples, which amplifies the ability of the model to generalise across tasks. Each training task is referred to as an episode. Few-shot classification is a very important research field in few-shot learning. Accordingly, there are two branches of relevant prior work: optimisation-based methods and metric-based methods.

2.1. Optimisation-based Methods

Optimisation-based methods[11, 12, 28, 35] aim to learn better initial parameters, thereby allowing new tasks to adapt more quickly. An LSTM meta-learner[28] generated a finer optimiser for another network by replacing the stochastic gradient descent with memory. Whereas the LSTM meta-learner involved two models, MAML[11], a typical algorithm, features only one model to optimise novel tasks using a few gradient descent steps. Subsequent works have made improvements in these networks. For instance, TAML[17] focused on learning an unbiased initial model to avoid excessive bias toward existing tasks. Further, MTL[35] combined the advantages of meta-learning and transfer-learning to provide an appropriate initialisation for the model. Gidaris and Komodakis[12] utilised the meta-learner network component to compose a weight generator that learns features with prior knowledge for few-shot classification. However, all the aforementioned methods involve bi-level optimisation and require fine-tuning on new tasks. In contrast, our method is end-to-end and does not require parameter updates on new samples to adapt to novel tasks.

2.2. Metric-based Methods

Metric-based methods predict class labels by computing the similarity between samples. Different similarity measurements have led to several few-shot classification frameworks. Matching Network[40] utilised cosine similarity and attention kernels to predict query class labels. Relation Network[36] employed CNN to compute the similarity. Prototypical Network[34] treated the computed mean as the prototype of each class and measured the Euclidean distance for query samples. We selected ProtoNet, which has been widely adopted in other research studies[29, 38], as our baseline. However, the basic frameworks mentioned above overlook the influence of appearance variations in congenetic samples. To address this, DeepEMD[48] introduced the Earth-Mover distance[30] to compute optimal matching flows for determining image classes and design a cross-reference mechanism to minimise the impact of appearance. Although effective, this approach has a significant computational burden. In contrast, our approach, HF-extractor, is concise and prevents models from neglecting the overall object identification goal, thus mitigating the impact of appearance variations.

On the one hand, previous works have focused on obtaining features that contain more useful information. FRN[44] introduced the concept of reconstruction and regression from support features to query features to make congenetic

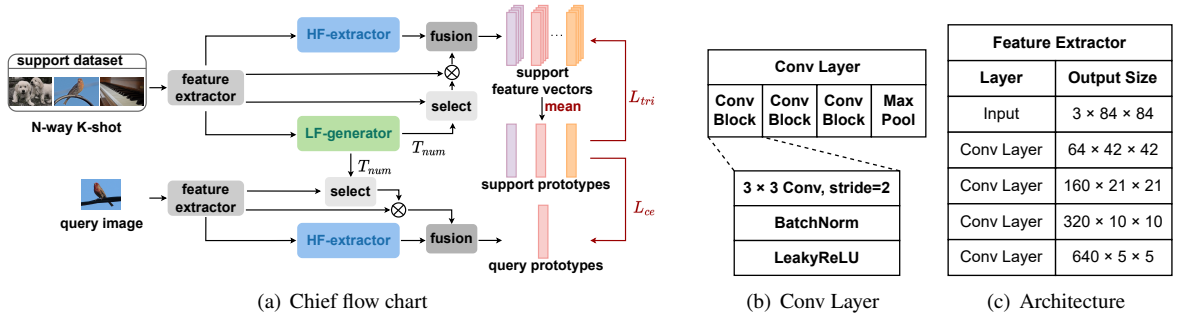


Figure 2: Chief flow chart (a), Convolution Layer (b), and Architecture of feature extractor (c). Here, (a) describes the general flow of our approach. Note that L-activators of the query are selected according to the same support activator numbers T_{num} . L_{ce} is generated between the support set and query set, whereas L_{tri} is calculated at the prototype level.

features more similar. CAN[15] explored the relationship between support and query features using a cross-attention module. Nevertheless, they all tend to explore the correlation of samples between support and query sets, which may introduce noise from the inter-set. Our LF-generator differentiates samples within the same set to explore the intrinsic class-related semantic information contained with them by unsupervised strategy.

On the other hand, some studies have investigated the loss function. Neg-Cosine[23] achieved excellent performance as an all-purpose loss. K-tuplet loss[21] aimed to gather similar samples close and separate different ones. Unlike these methods, which expect to enhance the generalisation of novel class features or learn discriminative features, our method delves deeper into the essence and seeks to obtain high-quality prototypes to improve the few-shot classification. In this regard, we redefine the sampling strategy of the Triplet loss to acquire better prototypes.

3. Method

The chief flow chart is shown in Fig. 2(a). The support and query samples are fed into the consistent feature extractor, followed by two streams to extract complementary semantic information. Specifically, given the support samples, the LF-generator learns the activator number T_{num} associated with discriminative semantic details. The task of T_{num} is to be shared in the query and support samples to strengthen the interaction between the support set and query set, where T_{num} represents the channel numbers corresponding to the sorted variance in descending order. In synchronization with the LF-generator, an HF-extractor delves into holistic semantic information. To obtain high-quality prototypes to facilitate subsequent classification, we not only calculate the inter-set cross-entropy loss L_{ce} but also consider the auxiliary loss L_{tri} between prototypes.

In this section, we first define some notations involved in the problem. This is followed by a detailed description of the operation mechanism of MIFN, consisting of an LF-generator and an HF-extractor. Finally, we incorporate triplet loss to aid few-shot classification at the prototype level.

3.1. Preliminary

We define an image and its corresponding class label as (x, y) . $D_b = \{(x_i, y_i)\}_{i=1}^{E_b}$ is the base dataset for the pre-training model to learn appropriate representations. In accordance with few-shot learning, we intend to predict the labels of samples in the query set $D_q = \{(x_i, y_i)\}_{i=1}^{E_q}$ given the support set $D_s = \{(x_i, y_i)\}_{i=1}^{E_s}$. Since there are N classes in D_s and K samples in each class, we refer to this episode as an N-way K-shot. Note that the novel classes C_n grouped by the support and query samples are disjoint from base classes C_b , i.e., $C_n \cap C_b = \emptyset$. Furthermore, we regard D_q as unlabelled samples because the corresponding labels are used for validation. Moreover, we treat the feature extractor as f_θ trained on the base dataset with the traditional cross-entropy loss function.

We consider not only the general few-shot classification but also the cross-domain scenario in which the training set and test set are from different domains.

3.2. Multisemantic Information Fusion Network

MIFN aims to carry more valuable semantic information into the limited representation space of the feature vectors and mitigate the influence of appearance variations. An overview is provided in Fig. 3, where the outputs of the feature

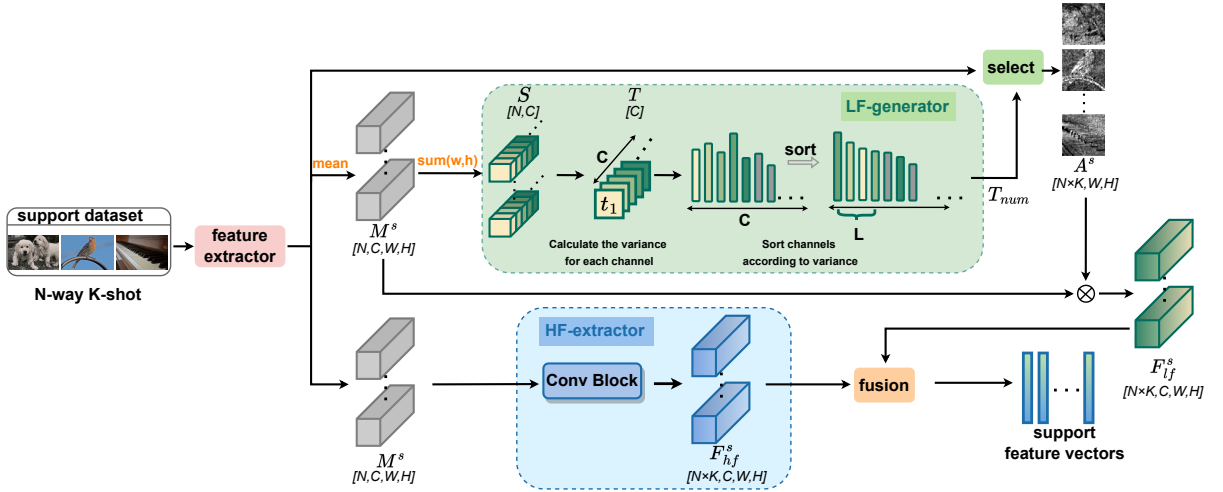


Figure 3: Overview of MIFN for few-shot classification. MIFN consists of two streams, LF-generator and HF-extractor. The support feature maps are initially processed through the feature extractor and then fed into the LF-generator to obtain activator numbers denoted as T_{num} . These T_{num} serve as guidance for selecting and activating discriminative semantic information from the support and query feature maps. The activated information is then summed element-wise with the global semantic information learned by the HF-extractor, resulting in the final feature vectors.

extractor flow to the respective streams (LF-generator and HF-extractor). These branches learn discriminative semantic details and holistic semantic information of the overall object, which are then aggregated into the final representations.

3.2.1. LF-generator

We propose an LF-generator that implicitly highlights class-related regions to generate more responsive semantic details without label supervision. The output of the support samples x^s through f_θ are feature maps $F^s = f_\theta(x^s) \in \mathbb{R}^{N \times K, C, W, H}$. The mean feature maps of each class $M^s \in \mathbb{R}^{N, C, W, H}$ are fed into the subsequent network to increase the otherness of the category.

First, we describe the summing descriptors $S \in \mathbb{R}^{N, C}$, which represent the aggregated semantic content of each channel under different classes, by sum pooling the M^s .

$$S_i = \sum_{w=0}^W \sum_{h=0}^H M_i^s(w, h) \quad (1)$$

Feature maps M^s through the spatial dimension shrink to channel descriptors with all information. These aggregated descriptors allow the subsequent calculation of channel variances and reflect global class-related channel information. The responses of each summing vector to the corresponding class deduce the difference of the same channel between classes. We compute the variances $T = \{t_1, t_2, \dots, t_c\}$ to represent this discriminability.

$$T = \sum_{i=1}^N (S_i - \frac{1}{N} \sum_{j=1}^N S_j)^2 \quad (2)$$

We aim to assign class-related weight to each channel through the variances, which reflect the discrepancy between the samples in the same channel. Channels with higher weights exhibit stronger responses to specific objects, thus possessing more discriminative semantic information.

Next, we select L channel indices with class-related semantic information by sorting the variances T corresponding to each channel, denoted as activator numbers T_{num} . We then index the corresponding layers as specific L-activators $A^s \in \mathbb{R}^{N \times K, W, H}$ for each sample. The L ablation is described in Section 5.1. The selected L-activators have a high response to discriminative semantic content. Consequently, these facilitate the learning of classified information and suppress uncorrelated noise. Moreover, because the activators are generated in separate support sets, we note the class

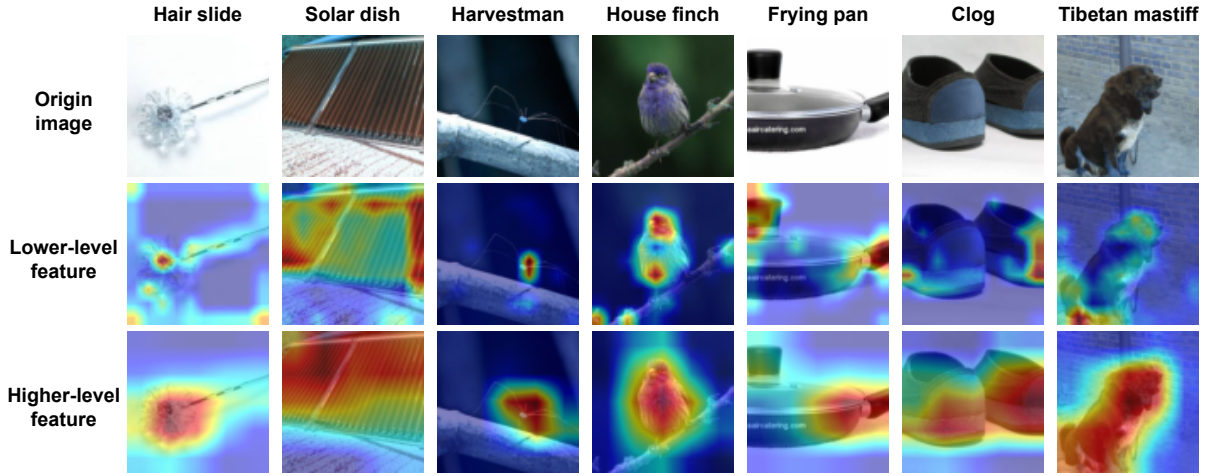


Figure 4: Illustration of the higher-level features with human cognition. **Top:** Query images. **Middle:** Class activation mapping of lower-level features. **Bottom:** Class activation mapping of higher-level features. Higher-level features can comprise most of the target area and capture global semantic information.

discrepancy itself and ignore the potential differential effects of the inter-set.

$$A_i^s = \sum_l^L M_i^s[T_{num}[l]], i = 1, 2, \dots, N \times K \quad (3)$$

Similar to the above operations, the query features also generate L-activators A^q based on the activator numbers T_{num} selected from the support set. We utilise the same values T_{num} to constrain the query features, enabling them to embed semantic details from the support features. This implementation implicitly shares the support knowledge over the query samples and closes the distance of the congeneric feature within the inter-set.

We activate the responsive class semantic information of features with weights $W \in \mathbb{R}^{N \times K, W, H}$ formed by L-activators A .

$$W^s = \mathbb{P}\left(\frac{A^s}{\sum_w^W \sum_h^H A^s(w, h)}\right) \\ W^q = \alpha \mathbb{G}(W^s) + \beta \mathbb{P}\left(\frac{A^q}{\sum_w^W \sum_h^H A^q(w, h)}\right) \quad (4)$$

where $\mathbb{P}(\cdot)$ is the sigmoid function and $\mathbb{G}(\cdot)$ is the repeat operator limiting the same dimension of the operand. The parameters for power-scaling are set as $\alpha = 0.5$ and $\beta = 1.0$, whose values are described in Section 5.1. Query weight W^q also imports support activators A^s so that the query features absorb the class-related semantic details from the support set, passing the congenetic information between different sets. We weight the initial feature maps to obtain the novel feature maps F_{lf} of the activated discriminative region.

$$F_{lf}^* = W^* F^* \quad (5)$$

where $*$ represents the corresponding feature map of support or query samples.

3.2.2. HF-extractor

LF-generator fully considers class-related discriminative semantic details that tend toward the local region. However, the prediction of classes also considers the global appearance, which approaches the human perception of the main object. The overall perspective facilitates a better understanding of samples from the same class with large differences. For example, a zebra crossing is more similar to a zebra from a different class and more discrepant from a horse from the same broad species. In this scenario, relying solely on local discriminative information can

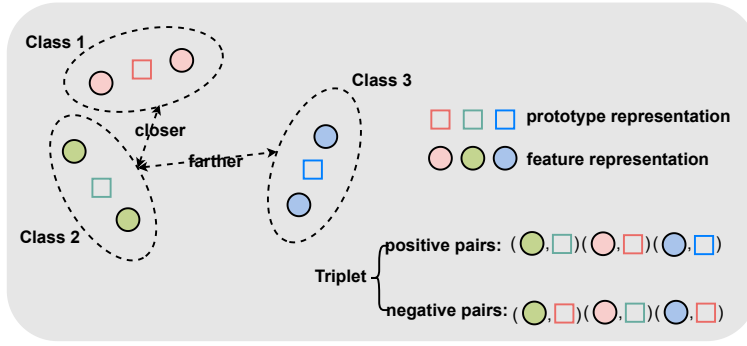


Figure 5: Demonstration of positive and negative pairs in the triplet loss.

easily lead to confusion. HF-extractor introduces a deeper convolutional block, as shown in Fig. 2(b), to stockpile higher-level features with comprehensive semantic details. As shown in Fig. 4, higher-level features, in contrast to lower-level features that emphasise discriminative details, cover a broader area of the objects. This indicates that higher-level features can reflect the overall appearance of the object and align better with the human cognition of the targets, thereby mitigating the impact of variations and providing effective guidance for classification. Unlike the LF-generator, which treats the support feature maps F^s as a single input, HF-extractor handles F^s and F^q . We denote the output of this module with softmax layers as F_{hf} . We discuss in Section 5.1 whether the quantity of convolutional blocks in HF-extractor has different impacts on few-shot classification.

3.2.3. Feature fusion module

LF-generator activates discriminative semantic details by selecting channels with significant class distinctions, while HF-extractor introduces higher-level semantic content mirroring human cognition of the objects. The rough outputs of these two branches mutually refine and complement to form a more finer multisemantic information, facilitating a more consolidated understanding of the classified targets. We perform an element-wise summation of the outputs from the above two modules to fuse semantic information from multiple perspectives.

$$\hat{F} = \lambda_1 F_{lf} + \lambda_2 F_{hf} \quad (6)$$

where λ_1 and λ_2 are the power-scaling parameters. The specific parameters are analysed in Section 5.1. The aggregated feature maps are converted via global average pooling to obtain the final feature vectors. It should be noted that MIFN is so simple that it can be directly inserted into ProtoNet. In addition, informative feature vectors can be obtained without requiring too many parameters or significant memory consumption.

3.3. Auxiliary loss

We adopt a triplet frame to learn more distinct prototypes and optimise the classification based on the prototype. Unlike traditional loss, triplet loss acquires discriminative embedding without the supervision of the concrete class and depends on the selection of positive pairs $\{F_a, F_p\}$ and negative pairs $\{F_a, F_n\}$.

$$L_{tri} = \max(\|\hat{F}_a - \hat{F}_p\|_2^2 - \|\hat{F}_a - \hat{F}_n\|_2^2 + \kappa, 0) \quad (7)$$

where κ is the margin for modifying the separability.

We reformulate the sampling strategy, as depicted in Fig. 5, to approach the selection process of positive and negative pairs for few-shot classification. For support samples serving as anchors, the corresponding prototypes naturally comprise positive pairs. The consideration is the form of the negative pairs. From the negative pairs comprised of heterologous samples, we treat prototypes with the maximum similarity to the class of the relevant samples as negative examples. In other words, the more indistinguishable samples are more faithful than the others. The margin κ guides the network to learn the embedding space where more congenetic features are closer together and features of different classes are farther apart. κ is not a fixed parameter but an adaptively selected value that represents the maximum distance between the class prototypes. Auxiliary loss aims to encourage feature vectors from the same

Model	Backbone	<i>mini</i> -ImageNet	
		1-shot	5-shot
ProtoNet[34, 46]	ResNet-12	62.39±0.21	80.53±0.14
MatchNet[40, 46, 48]	ResNet-12	65.64±0.20	78.72±0.15
MetaOptNet[20]	ResNet-12	62.64±0.61	78.63±0.46
CAN[15]	ResNet-12	63.85±0.48	79.44±0.34
Meta-Baseline[5]	ResNet-12	63.17±0.23	79.26±0.17
SimpleShot[43]	ResNet-18	62.85±0.20	80.02±0.14
Robust20-distill[8]	ResNet-18	63.06±0.61	80.63±0.42
S2M2[26]	ResNet-18	64.06±0.18	80.58±0.12
DSN[33]	ResNet-12	62.64±0.66	78.83±0.45
Neg-Cosine[23]	ResNet-12	63.85±0.81	81.57±0.56
Afrasiyabi et al.[1]	ResNet-18	59.88±0.67	80.35±0.73
E^3 BM[24]	ResNet-12	64.09±0.37	80.29±0.25
DeepEMD[48]	ResNet-12	65.91±0.82	82.41±0.56
RFS-simple[37]	ResNet-12	62.02±0.63	79.64±0.44
LASTSHOT[47]	ResNet-12	64.80±0.20	81.65±0.14
P-transfer[32]	ResNet-12	64.21±0.77	80.38±0.59
Mini800_SSL[3]	AmdimNet	64.03±0.20	81.51±0.14
HyperProto[7]	ResNet-12	59.65±0.62	78.24±0.47
QSFormer [42]	ResNet-12	65.24±0.28	79.96±0.20
MIFN(<i>ours</i>)	ResNet-12	65.42±0.65	80.85±0.41
MIFN(<i>ours</i>)+ L_{tri}	ResNet-12	66.43±0.63	81.51±0.42

Table 1

Performance of competitive few-shot methods on *mini*-ImageNet.

classes to be clustered together in the same manifold, while ensuring that features of different classes are pushed far apart, thus sharpening the decision boundary between prototypes and obtaining high-quality prototypes.

In summary, the final loss function consists of the traditional cross-entropy loss L_{ce} and the proposed auxiliary loss L_{tri} .

$$L = L_{ce} + \gamma L_{tri} \quad (8)$$

where γ is the fixed parameter of power-scaling with 0.1. Note that our method is end-to-end and does not require multi-stage training to obtain the results quickly and easily.

4. Experiment

MIFN incorporates the overall semantic information in the limited embedding space to grasp the all-sided understanding of the objects. We conducted extensive experiments to demonstrate that the proposed method is competitive in both general and cross-domain few-shot classification.

Implementation details: We first train the feature extractor with a cross-entropy loss on the base classes. For a fair comparison, we utilise ResNet-12[14] as the feature extractor containing four convolutional layers, as shown in Fig. 2(c). The introduction of pre-training effectively boosts performance. During the meta-training, we adopt ProtoNet as our baseline and the SGD optimiser[50] with the fixed learning rate of $3e - 4$. For the final results, we evaluate the 5-way 1-shot and 5-way 5-shot settings on 600 random episodes with 15 query samples.

4.1. General Few-shot Classification

We evaluate the performance of our method under different settings on two common datasets, *mini*-ImageNet and *tiered*-ImageNet.

Mini-ImageNet is an excerpt from ImageNet[31] and contains 60000 images of size 84×84 in 100 categories, with 600 samples per class. For a fair comparison, we split this dataset into 64 training, 16 validation, and 20 test classes.

Model	Backbone	<i>tiered-ImageNet</i>	
		1-shot	5-shot
MatchNet[40, 46, 48]	ResNet-12	68.50±0.92	80.60±0.71
ProtoNet[34, 46]	ResNet-12	68.23±0.23	84.03±0.16
MetaOptNet[20]	ResNet-12	65.99±0.72	81.56±0.53
Robust20-distill[8]	ResNet-18	65.43±0.21	70.44±0.32
SimpleShot[43]	ResNet-18	69.09±0.22	84.58±0.16
CAN[15]	ResNet-12	69.89±0.51	84.23±0.37
Meta-Baseline[5]	ResNet-12	68.62±0.27	83.29±0.18
DSN[33]	ResNet-12	66.22±0.75	82.79±0.48
Afrasiyabi et al.[1]	ResNet-18	69.29±0.56	85.97±0.49
RFS-simple[37]	ResNet-12	69.74±0.72	84.41±0.55
LASTSHOT[47]	ResNet-12	69.37±0.23	85.36±0.16
SENet [49]	ResNet-12	64.88±0.72	81.20±0.55
MIFN(<i>ours</i>)	ResNet-12	69.54±0.67	84.04±0.49
MIFN(<i>ours</i>)+ L_{lri}	ResNet-12	70.03±0.72	84.14±0.50

Table 2

Performance of competitive few-shot methods on *tiered-ImageNet*.

Tiered-ImageNet is a larger subset of ImageNet with 351 training, 97 validation, and 160 test classes, totaling 779165 images of size 84×84 .

As shown in Tables 1 and 2, our method is still competitive against the recent State Of The Art (SOTA) approaches. On *mini-ImageNet*, MIFN alone, compared to the ProtoNet used as the baseline, increases performance by 3.03% in 1-shot and 0.3% in 5-shot. And our overall method is enhanced by 4.04% and 0.9%, respectively. On *tiered-ImageNet*, the improvements exceed 1.8% in 1-shot and 0.5% in 5-shot. Furthermore, we notice that our method exhibits a performance gap compared to DeepEMD in the 5-shot setting of *mini-ImageNet*. DeepEMD divides the image into regions and computes the optimal matching flow to represent the final distance, which assists to suppress background noise and improve performance. However, implementing this method requires DeepEMD to utilize qpth[2] for solving involved linear programming problems and computing gradients, consuming a substantial computation time. In contrast, our method only introduces an HF-extractor, which is more concise and computationally inexpensive, as evident in Table 9. In *tiered-ImageNet*, our method exhibits minimal performance differences compared to Afrasiyabi et al. and LASTSHOT. First, the discrepancy in the backbones used may contribute to performance differences. Afrasiyabi et al. used ResNet-18, whereas we opted for the more commonly employed ResNet-12. Second, we refrained from incorporating any additional measures beyond pre-training. Afrasiyabi et al. introduced novel class samples for fine-tuning in a two-stage training process, which inherently involves more intricate computations. LASTSHOT adopted knowledge distillation, an additional technology that typically improves performance to some extent. For fair comparisons, we did not introduce knowledge distillation. Furthermore, our technique does not require adjustment of parameters in the inference stage and is plug-and-play, directly embedding into ProtoNet.

4.2. Cross-domain Few-shot Classification

We consider the domain shift in the few-shot classification where models trained on *mini-ImageNet* are evaluated on other datasets. We aim to test the generalisation of our method and its effectiveness in addressing the domain gap.

CUB is a fine-grained dataset containing 11788 images in 200 bird subcategories. We randomly divided the data into 100 training, 50 validation, and 50 test classes with the same ratio as most methods.

Caltech-101 contains 101 object classes and a background class for a total of 9146 images. We randomly selected 21 classes as test classes according to the *mini-ImageNet* ratio.

Stanford-dogs is also a fine-grained dataset with 120 dog subcategories, approximately 150 samples per class, and a total of 20580 images. Following the CUB classification ratio, we randomly split the dataset into 72 training, 24 validation, and 24 test classes.

Stanford-cars is primarily used for fine-grained classification tasks. This dataset contains 16185 images and 196 classes. We randomly divided Stanford-cars into 118 training, 39 validation, and 39 test classes.

Model	Backbone	CUB	
		1-shot	5-shot
MAML[11, 4]	ResNet-18	-	51.34±0.72
ProtoNet[34, 46]	ResNet-18	-	62.02±0.70
Baseline[4]	ResNet-18	-	65.57±0.70
Diverse 20-full[8]	ResNet-18	-	66.17±0.55
SimpleShot[43]	ResNet-18	48.56	65.63
MetaOptNet[20]	ResNet-12	44.79±0.75	64.98±0.68
MatchNet+FT[39]	ResNet-10	36.61±0.53	55.23±0.83
RelationNet+FT[39]	ResNet-10	44.07±0.77	59.46±0.71
GNN+FT[39]	ResNet-10	47.47±0.75	66.98±0.68
MN+AFA [16]	ResNet-10	41.02	59.46
MIFN(<i>ours</i>)	ResNet-12	48.21±0.60	65.33±0.54
MIFN(<i>ours</i>)+ L_{tri}	ResNet-12	49.21±0.60	66.30±0.56

Table 3

Performance comparison in the cross-domain setting: *mini*-ImageNet→CUB. '-': not reported.

	Model	Caltech-101	Stanford-dogs	Stanford-cars
5-way 1-shot	ProtoNet[34, 46]	73.39±0.65	51.32±0.66	32.99±0.45
	MIFN(<i>ours</i>)+ L_{tri}	78.24±0.60	54.37±0.68	33.82±0.46
5-way 5-shot	ProtoNet[34, 46]	90.03±0.33	70.99±0.54	47.71±0.52
	MIFN(<i>ours</i>)+ L_{tri}	91.30±0.34	71.33±0.55	48.47±0.52

Table 4

Performance comparison in the cross-domain setting: *mini*-ImageNet→ Other datasets.

We directly utilise a model trained on *mini*-ImageNet to test the performance of the novel dataset. Note that there is no fine-tuning or data augmentation[4, 43, 46, 48], such as random cropping, right-left flipping, color jitter. The results of the cross-domain scenario are presented in Table 3 and Table 4. The advantages of the proposed method are obvious. For Caltech-101, Stanford-dogs, and Stanford-cars, our method exhibits an improvement of 1 to 5 accuracy points in 1-shot settings and approximately 1 point in 5-shot settings compared to ProtoNet. The accuracy on Caltech-101 is considerably high, even exceeding the accuracy of testing after training on *mini*-ImageNet, we suspect that the images in this dataset eliminate the impact of background, learning more pure information. In addition, for CUB, our method exhibits excellent performance compared with other recent methods. Furthermore, MIFN alone can achieve a 3.31% improvement on CUB. From the above experiments, we conclude that MIFN is effective for the cross-domain problem and practical for fine-grained few-shot classification.

5. Analysis

5.1. Ablation experiment

We further discuss MIFN to elaborate on the effectiveness of each part of the proposed method.

Effectiveness of each part of the method: MIFN consists of two branches, namely HF-extractor and LF-generator. We conduct the ablation experiments with a 5-way 1-shot setting to analyse the effectiveness of each part. As shown in Fig. 6, each stream improves the baseline to a different degree, i.e., each stream enables feature vectors to introduce useful information of different utility. All components improve the baseline to some extent on *mini*-ImageNet, contributing to an increase of 2-3 points respectively. Moreover, the fused representations perform better than individual outputs. This demonstrates that each additional part positively influences the baseline and multisemantic information potentially inspires object recognition. Surprisingly, HF-extractor does not demonstrate superior performance on *tiered*-ImageNet. We speculate that this could be because *tiered*-ImageNet is divided into more nuanced classes compared to *mini*-ImageNet. Thus, the model needs to capture more subtle details for distinction. However, the HF-extractor, which prefers to extract global semantic information, may not be suitable for the abstraction of subtle details and necessitates the support of the LF-generator to learn finer distinctions, and the two complement

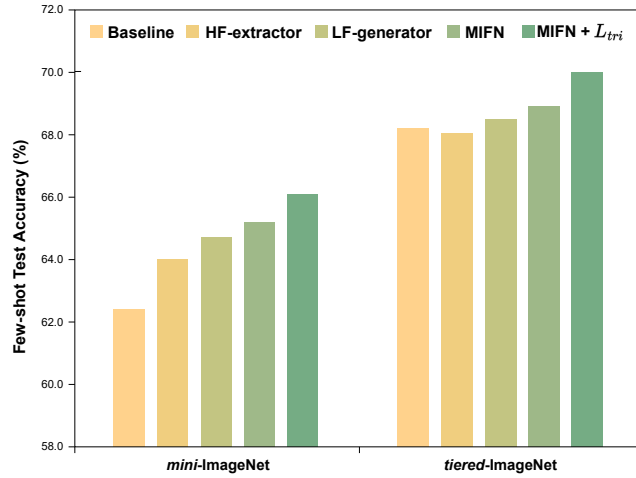


Figure 6: The accuracy of our method with different components on *mini-ImageNet* and *tiered-ImageNet* with 5-way 1-shot.

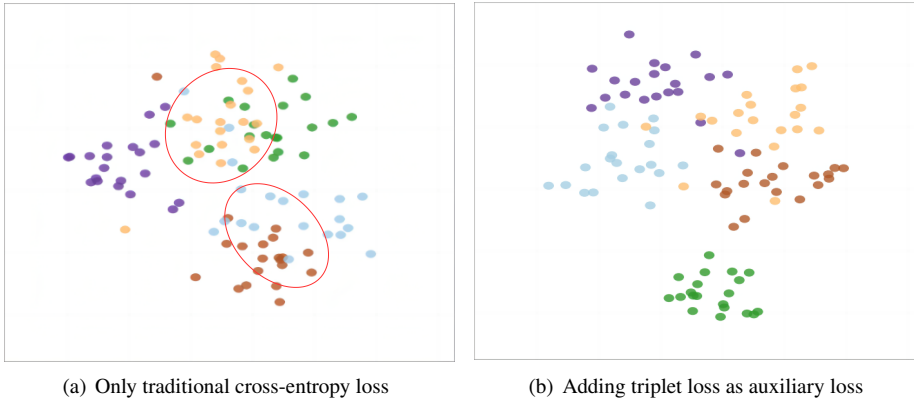


Figure 7: Visualization of support samples in ProtoNet with different loss functions. The red circle is the class confusion area.

each other. Moreover, MIFN is equipped with auxiliary loss support to achieve better results. This illustrates that sharpening decision boundaries at the prototype level facilitates subsequent classification operations.

Auxiliary loss: To further verify the effectiveness of the auxiliary loss, we conduct a comparative test on ProtoNet without the influence of other methods. We use a 5-way 20-shot setting instead of a 5-way 1-shot to highlight variations at the prototype level. T-SNE[25] is used to visualise the distribution of support samples. As shown in Fig.7, the boundaries between classes in the red box are blurred in the left figure, which is caused by traditional cross-entropy loss training alone. By contrast, the right image that uses an auxiliary loss exhibits the sharper decision boundary and the obvious clustering effect. We confirm that auxiliary loss enhances separability at the prototype level, such that the category predictions based on the prototypes are more accurate.

Number of L: We select L channels of the feature map corresponding to maximal variances, introducing discriminative information into the feature vector in the LF-generator. As shown in Table 5, we achieve good performance with 5-way 1-shot when $L=25$ and when $L=5$ in 5-way 5-shot. The results demonstrate that the number of L is not proportional to the performance, and channels with maximal variances are valuable for classification.

The performance of support and query activators in different proportions: In the LF-generator, the variation in the proportion of semantic information between the support and query samples has a discernible impact on performance. As shown in Table 6, we observe that the best performance is achieved when $\alpha=0.5$ and $\beta=1.0$. Query samples absorb an appropriate amount of support semantic information, facilitating effective knowledge exchange

Number of L	<i>mini</i> -ImageNet	
	5-way 1-shot	5-way 5-shot
L=5	65.73±0.64	81.51±0.42
L=25	66.43±0.63	81.43±0.41
L=50	65.46±0.62	81.06±0.45
L=75	65.73±0.63	81.33±0.44
L=100	65.90±0.63	81.28±0.41

Table 5

Performance of different number of selected activators (L) on *mini*-ImageNet.

α	β	<i>mini</i> -ImageNet		<i>tiered</i> -ImageNet	
		5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
0.5	0.5	64.29±0.64	80.53±0.42	65.88±0.75	82.48±0.50
0.5	1.0	66.43±0.63	81.51±0.42	70.03±0.72	84.14±0.50
1.0	0.5	66.50±0.62	79.54±0.43	68.31±0.73	82.54±0.52
1.0	1.0	65.71±0.62	80.24±0.42	69.06±0.66	81.39±0.52

Table 6

Performance of support and query activators in different proportions on *mini*-ImageNet and *tiered*-ImageNet.

		5-way 1-shot			5-way 5-shot		
		$\lambda_2 = 0.0$	$\lambda_2 = 0.5$	$\lambda_2 = 1.0$	$\lambda_2 = 0.0$	$\lambda_2 = 0.5$	$\lambda_2 = 1.0$
$\lambda_1 = 0.0$	-	64.53±0.65	64.36±0.65		-	80.51±0.45	80.54±0.45
$\lambda_1 = 0.5$	64.56±0.64	66.43±0.63	65.15±0.64		80.23±0.47	81.51±0.42	80.61±0.46
$\lambda_1 = 1.0$	64.90±0.64	65.75±0.64	65.48±0.64		80.30±0.47	80.52±0.47	80.61±0.46

Table 7

Performance of different parameter values on *mini*-ImageNet with 5-way 1-shot and 5-way 5-shot. '-': meaningless.

HF-extractor	Memory	Acc
1	5382	66.43±0.63
2	5414	65.44±0.65
3	5430	65.71±0.65

Table 8

Performance of different number of Conv blocks in the HF-extractor.

between sets without introducing inter-set noise. We suspect that if the proportion of support semantic information becomes excessively large, the query sample may exhibit a higher inclination towards resembling the support sets, compromising its own distinctiveness.

The proportion requirement for two types of semantic information: We explore whether few-shot classification benefits more from discriminative or macro semantic information. The parameter values λ_1 and λ_2 determine different proportions of corresponding semantic information in the final features. We conduct the experiments using *mini*-ImageNet. As shown in Table 7, the performance is optimal in a 5-way 1-shot setting when $\lambda_1 = 0.5$ and $\lambda_2 = 0.5$ and the similar performance is observed in a 5-way 5-shot setting. This indicates that few-shot classification has a balanced demand for these two types of semantic information. Discriminative semantic details aim to extract representative features of different categories. However, these details may ignore the consistent information of samples of the same class, which requires macro semantic information to compensate. We suspect that the multisemantic information implicitly sharpens the manifold regions of different categories while restricting the placement of congenetic samples in the same manifold.

The composition of HF-extractor: We extract higher-level semantic information using an HF-extractor, with the expectation of introducing more human cognition into the feature vectors, enabling a more holistic understanding of the objects. We explore whether adding more computational blocks to the HF-extractor contributes to the extraction of

Model	Phase	Episode	Time per Episode(ms)
DeepEMD[48]	meta-train	5-way 1-shot	26,000
ProtoNet[34, 46]	meta-train	5-way 1-shot	210
MIFN(<i>ours</i>)	meta-train	5-way 1-shot	300
DeepEMD[48]	meta-train	5-way 5-shot	>990,000
ProtoNet[34, 46]	meta-train	5-way 5-shot	250
MIFN(<i>ours</i>)	meta-train	5-way 5-shot	270
DeepEMD[48]	meta-test	5-way 1-shot	23,275
ProtoNet[34, 46]	meta-test	5-way 1-shot	83
MIFN(<i>ours</i>)	meta-test	5-way 1-shot	147
DeepEMD[48]	meta-test	5-way 5-shot	>800,000
ProtoNet[34, 46]	meta-test	5-way 5-shot	105
MIFN(<i>ours</i>)	meta-test	5-way 5-shot	137

Table 9

Speed comparison between ProtoNet, DeepEMD and MIFN on *mini*-ImageNet with Resnet-12 as the backbone.

Model	5-way 1-shot	5-way 5-shot
ProtoNet[34, 46]	5296	6140
MIFN(<i>ours</i>)	5382	6754

Table 10

Memory comparison between ProtoNet and MIFN on *mini*-ImageNet with Resnet-12 as the backbone.

human cognition. As shown in Table 8, with the deepening of the HF-extractor, the model does not achieve the expected performance. We suspect that the generated over-higher-level semantic information neutralises local discriminative details learned from the LF-generator to block the expression of the final feature vector.

5.2. Computational efficiency

Speed comparison: MIFN can be directly embedded into ProtoNet for end-to-end training. We achieve the expected performance without adopting additional parameters. As shown in Table 9, our method has a significant acceleration compared to DeepEMD and does not show an evident decline in relation to the baseline in different settings. This indicates that MIFN is sufficiently concise.

Memory comparison: MIFN does not introduce many parameters or modules into the model to boost performance. To support this assertion, we compare MIFN with ProtoNet on GPU usage in the training process. We use the **nvidia-smi** command to report the GPU usage. As shown in Table 10, for 84×84 images, the memory consumption of our method is close to the baseline on *mini*-ImageNet. Our two-stream model is effective without bells and whistles.

6. Conclusion

In this paper, we propose MIFN seamlessly embedded in ProtoNet, stockpiling consolidated semantic information in the limited space of feature vectors and overcoming the appearance variations. Here, the LF-generator activates class-related discriminative semantic details using an unsupervised mechanism, and the HF-extractor introduces higher-level semantic information mimicking human cognition. The outputs of the two modules are complementary, enhancing the separability of features from different classes and maintaining the consistency of congenetic samples. Further, we incorporate triplet loss with a redefined sampling strategy to sharpen the decision boundary at the prototype level to support subsequent classification. We demonstrate that our method is competitive on general few-shot classification and cross-domain problems, and the effectiveness of each stream is evaluated by ablation experiments.

Declaration of Competing Interest

The authors declare that they do not have any financial or nonfinancial conflict of interests.

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